

Block chaining: what runs before what, and on which file

Mainland Portugal, 24 NUTS III regions, 2025-2040. Blocks are run by hand in numerical order; the arrows are the contracts between them.

data layer
(00_data)

master_panel_nuts3.xlsx

script 10

master_panel_nuts3_with_age.xlsx

script 11

teacher_supply_panel.xlsx

script 06

read by 04 directly:
it never enters
the master panel

model
blocks

01_students

enrolment by
NUTS III and cycle

02_demand

teaching posts
implied by enrolment

03_supply

workforce, cohort chain
UNCONSTRAINED entry

04_gap

applies the ITE
restriction

05_uncertainty

Monte Carlo over
the 04 parameters

06_validation

tests the chain,
feeds none of it

students_forecast_nuts3.xlsx

teacher_demand_selected.csv

supply_projection.csv
supply_recruitment_by_channel.csv

pipeline_gap.csv
gap_national.csv

supply_projection.csv

gap_national.csv · gap_recruitment_nuts3.csv

recruitment_montecarlo_v3.csv (the demand sigma)

teacher_demand_selected.csv

06_validation opens the outputs of every preceding block and tests them. Nothing it writes is read by any block, so a failure there cannot alter a result upstream.

The forward chain's arithmetic closes. Largest absolute residuals reported by 03_supply: flow identity 2.9e-11, historical gross flows 9.1e-13, hazard times stock against gross exits 1.1e-13, band-fill helper exactly zero. Acyclicity is what makes those identities checkable in a single pass.

— value flow: a quantity the next block computes with — value flow from the data layer - - - BACKWARD read: opened for consistency checks only terminal read: 06 tests, and feeds nothing